



National Comprehensive
Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Dermatofibrosarcoma Protuberans

Version 2.2026 — October 24, 2025

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**NCCN recognizes the importance of clinical trials and encourages participation when applicable and available.
Trials should be designed to maximize inclusiveness and broad representative enrollment.**

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National Comprehensive
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NCCN肿瘤学临床实践指南（NCCN指南®）

隆起性皮肤纤维肉瘤

版本 2.2026 — 2025 年 10 月 24 日

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NCCN认识到临床试验的重要性，并在适用且条件允许时鼓励参与。
试验设计应旨在最大限度地提高包容性并实现广泛的代表性入组。

继续



***Jeremy Bordeaux, MD, MPH/Chair** ☐
Case Comprehensive Cancer Center/
University Hospitals Seidman Cancer Center
and Cleveland Clinic Taussig Cancer Institute

***Paul Nghiem, MD, PhD/Vice Chair** ☐
Fred Hutchinson Cancer Center

Sumaira Z. Aasi, MD ☐
Stanford Cancer Institute

Murad Alam, MD, MBA, MSCI ☐ ☐ ☐
Robert H. Lurie Comprehensive Cancer
Center of Northwestern University

Arya Amini, MD §
City of Hope National Medical Center

Diana Bolotin, MD, PhD ☐
The UChicago Medicine
Comprehensive Cancer Center

Jennifer Chang, MD ☐
UC San Diego Moores Cancer Center

Pei-Ling Chen, MD, PhD ≠
Moffitt Cancer Center

Carlo M. Contreras, MD ☐
The Ohio State University Comprehensive
Cancer Center - James Cancer Hospital
and Solove Research Institute

Jessica M. Donigan, MD ☐ ☐
Huntsman Cancer Institute
at the University of Utah

Daniel Eisen, MD ☐
UC Davis Comprehensive Cancer Center

Jeffrey M. Farma, MD ☐
Fox Chase Cancer Center

*** Kelly Harms, MD, PhD** ☐
University of Michigan Rogel Cancer Center

Nicole LeBoeuf, MD, MPH ☐
Dana-Farber/Brigham and Women's Cancer
Center | Mass General Cancer Center

John Nicholas Lukens, MD §
Abramson Cancer Center
at the University of Pennsylvania

Susan Manber ≠
Publicis Health

Lawrence Mark, MD, PhD ☐
Indiana University Melvin and Bren Simon
Comprehensive Cancer Center

Stacy McMurray, MD ☐
Vanderbilt-Ingram Cancer Center

Theresa Medina, MD †
University of Colorado Cancer Center

Kishwer S. Nehal, MD ☐
Memorial Sloan Kettering Cancer Center

Kelly Olino, MD ☐
Yale Cancer Center/Smilow Cancer Hospital

Gyorgy Paragh, MD, PhD, FAAD ☐
Roswell Park Comprehensive Cancer Center

Soo Park, MD †
UC San Diego Moores Cancer Center

Tejesh Patel, MD ☐
The University of Tennessee Health Science Center

Jason Rich, MD §
Siteman Cancer Center at Barnes-Jewish Hospital
and Washington University School of Medicine

Ashok R. Shaha, MD ☐
Memorial Sloan Kettering Cancer Center

Bhavina Sharma, MD, MPH ‡
Fred & Pamela Buffett Cancer Center

Olayemi Sokumbi, MD ☐ ≠
Mayo Clinic Comprehensive Cancer Center

Divya Srivastava, MD ☐
UT Southwestern Simmons Comprehensive Cancer Center

Joel Sunshine, MD, PhD ☐ ≠
Johns Hopkins Kimmel Cancer Center

Valencia Thomas, MD ☐ ≠ ☐
The University of Texas MD Anderson Cancer Center

Puja Venkat, MD §
UCLA Jonsson Comprehensive Cancer Center

Melodi Whitley, MD, PhD ☐
Duke Cancer Institute

Yaohui Gloria Xu, MD, PhD ☐ ☐
University of Wisconsin
Carbone Cancer Center

Siegrid Yu, MD ☐
UCSF Helen Diller Family Comprehensive Cancer Center

Mehran Yusuf, MD §
O'Neal Comprehensive Cancer Center at UAB

NCCN

Sara Espinosa, PhD
Beth McCullough, RN, BS

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- *杰里米·博尔多，医学博士、公共卫生硕士/系主任 ω
案例：综合癌症中心/大学医院赛德曼癌症中心与克利夫兰诊所陶西格癌症研究所
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弗雷德·哈钦森癌症中心
- 苏迈拉·Z·阿西，医学博士 ω
斯坦福癌症研究所
- 穆拉德·阿拉姆，医学博士、工商管理硕士、理学硕士 ω $\Uparrow$ ζ
西北大学罗伯特·H·卢里综合癌症中心
- 阿里亚·阿米尼医学博士 $\S$
希望之城国家医疗中心
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芝加哥大学医学综合癌症中心
- Jennifer Chang，医学博士
加州大学圣地亚哥分校穆尔斯癌症中心
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加州大学戴维斯分校综合癌症中心
- 杰弗里·M·法玛医学博士 $\Uparrow$
福克斯蔡斯癌症中心

- * 凯莉·哈姆斯，医学博士、哲学博士 ω
密歇根大学罗格尔癌症中心
- 妮可·勒布夫，医学博士、公共卫生硕士 ω
丹娜-法伯/布莱根妇女癌症中心 | 马萨诸塞总医院癌症中心
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- 苏珊·曼伯 $\neq$
公共卫生
- 劳伦斯·马克，医学博士、哲学博士 ω
印第安纳大学梅尔文与布伦·西蒙综合癌症中心
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巴恩斯-犹太医院及华盛顿大学医学院下属的西特曼癌症中心

- 阿肖克·R·沙哈，医学博士 $\Uparrow$
斯隆-凯特琳癌症中心纪念医院
- 巴维娜·夏尔马，医学博士，公共卫生硕士 ‡
弗雷德与帕梅拉·巴菲特癌症中心
- Olayemi Sokumbi，医学博士 ω $\neq$
梅奥诊所综合癌症中心
- 迪维娅·斯里瓦斯塔瓦，医学博士 ω
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- Joel Sunshine，医学博士、哲学博士 ω $\neq$
约翰霍普金斯大学基梅尔癌症中心
- 瓦伦西亚·托马斯，医学博士 ω $\neq$ $\Uparrow$
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- 普贾·文卡特，医学博士 $\S$
加州大学洛杉矶分校琼森综合癌症中心
- 梅洛迪·惠特利，医学博士、哲学博士 ω
杜克癌症研究所
- 徐耀辉·格洛丽亚，医学博士、哲学博士 ω $\Uparrow$
威斯康星大学
卡博内癌症中心
- 徐格里德·余医学博士 ω
加州大学旧金山分校海伦·迪勒家族综合癌症中心
- 梅赫兰·优素福，医学博士 $\S$
阿拉巴马大学伯明翰分校奥尼尔综合癌症中心

NCCN
萨拉·埃斯皮诺萨，哲学博士
贝丝·麦卡洛，注册护士，理学学士

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[Summary of the Guidelines Updates](#)

[Clinical Presentation and Workup \(DFSP-1\)](#)
[Treatment \(DFSP-2\)](#)
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[Principles of Pathology \(DFSP-A\)](#)
[Principles of Surgery \(DFSP-B\)](#)
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[Abbreviations \(ABBR-1\)](#)

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See [NCCN Categories of Preference](#).

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[NCCN隆起性皮肤纤维肉瘤专家组成员指南更新摘要](#)

[临床表现与检查 \(DFSP-1\)](#)

[治疗 \(DFSP-2\)](#)

[复发/转移的随访与治疗 \(DFSP-3\)](#)

[病理学原理 \(DFSP-A\)](#)

[《外科手术 \(DFSP-B\)](#)

[放射治疗原理 \(DFSP-C\)](#)

[缩写 \(ABBR-1\)](#)

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Updates in Version 2.2026 of the NCCN Guidelines for Dermatofibrosarcoma Protuberans from Version 1.2026 include:

[MS-1](#)

- The discussion section has been updated to reflect changes in the algorithm.

Updates in Version 1.2026 of the NCCN Guidelines for Dermatofibrosarcoma Protuberans from Version 1.2025 include:

[DFSP-1](#)

- Preliminary Workup, second bullet, sub-bullet removed: Debulking specimens from all excisions should be examined to identify fibrosarcomatous transformation of dermatofibrosarcoma protuberans (FS-DFSP).
- Footnote removed: If areas of transformation to fibrosarcoma or other sarcoma subtypes are identified, multidisciplinary consultation for consideration of further treatment and surveillance is recommended. FS-DFSP is associated with a higher metastasis risk. The patient should be referred to a center with expertise in management of soft tissue sarcomas. See NCCN Guidelines for Soft Tissue Sarcoma for multimodal therapy and surveillance considerations including CT of draining nodal basin and chest.

[DFSP-2](#)

- Treatment, new bullet added: Debulking specimens from all excisions should be examined to identify fibrosarcomatous transformation of dermatofibrosarcoma protuberans (FS-DFSP).
- Adjuvant Treatment bottom pathway revised: ~~Multidisciplinary consultation for consideration of~~ *Consider* radiation therapy (RT) ~~or other therapy if margins are deemed narrow by treating physician and further resection would not be feasible upon recurrence.~~
- Follow-Up header and bullets in that column removed:
 - ▶ Physical exam with focus on primary site every 6–12 months.
 - ▶ Patient education about regular self-exam.
 - ▶ Consider MRI surveillance for deeply invasive disease or other concerns for recurrence.
- Last column revised: ~~Therapy for Recurrence/ Metastasis Follow-up~~ (DFSP-3).
- Footnotes revised:
 - ▶ Footnote c: . . . The patient should be referred to a center with expertise in management of soft tissue sarcomas. See NCCN Guidelines for Soft Tissue Sarcoma for multimodal therapy and surveillance considerations including CT of ~~draining nodal basin and~~ chest.
 - ▶ Footnote e: . . . Wide undermining is discouraged prior to confirmation of clear margins due to the difficulty of interpreting subsequent re-excised margins, and the risk of concealing residual tumor below mobilized tissue. See Principles of Surgery (DFSP-B). Curtis KK, et al. J Natl Compr Canc Netw 2024;22:e247036. (Also page DFSP-3)
- Footnote removed: When Mohs or other forms of PDEMA are not utilized and margins are negative, RT is not recommended. When Mohs or other forms of PDEMA are not utilized, consider RT if margins are considered narrow by the treating physicians. RT if not given previously and further resection is not feasible.

[Continued](#)

NCCN皮肤纤维肉瘤指南（版本1.2026）的2.2026版本更新内容包括：

MS-1

- 讨论部分已更新，以反映算法的变更。

NCCN皮肤纤维肉瘤指南（版本1.2025）升级至版本1.2026后的主要更新包括：

DFSP-1

- 初步检查结果（第二项要点，已删除子项）：所有切除标本均应进行减瘤病理检查，以明确隆起性皮肤纤维肉瘤（FS-DFSP）是否发生纤维肉瘤样转化。
- 脚注已删除：若发现病变区域向纤维肉瘤或其他肉瘤亚型转化，建议进行多学科会诊以评估进一步治疗方案及监测措施。FS-DFSP与较高的转移风险相关。患者应转诊至擅长软组织肉瘤治疗的专科中心。关于多模式治疗及监测方案（包括引流淋巴结区域及胸部CT检查），请参阅NCCN软组织肉瘤指南。

DFSP-2

- 新增治疗方案：所有切除标本均应进行减瘤病理检查，以明确隆起性皮肤纤维肉瘤（FS-DFSP）是否发生纤维肉瘤样转化。
- 辅助治疗路径修订如下：~~为评估病情进行多学科会诊；考虑放射治疗（RT）或其他治疗方案若主治医师判定切缘过窄，且复发后无法进行进一步切除术。~~
- 该列中的后续标题和项目符号已移除：
 - 每6至12个月进行一次以原发部位为重点的体格检查。
 - 关于定期自我检查的患者教育。
 - 建议采用MRI监测深部浸润性疾病或其他复发风险。
- 最后一列修订内容：~~复发/转移治疗随访（DFSP-3）。~~
- 脚注已修订：
 - 脚注c：该患者应转诊至擅长软组织肉瘤治疗的专科中心。关于多模式治疗及监测方案（包括排液淋巴结区域及胸部CT检查），请参阅NCCN软组织肉瘤指南。
 - 脚注e：在确认切缘清晰之前，不建议进行广泛的切缘扩大术，原因在于后续重新切除的切缘难以准确判读，且存在使游离组织下方残留肿瘤被掩盖的风险。详见《外科学原理》（DFSP-B）。Curtis KK 等人，《美国国家综合癌症网络杂志》2024；22:e247036。（另见 DFSP-3 页）
- 脚注已删除：当未使用Mohs或其他形式的PDEMA且切缘呈阴性时，不推荐采用放射治疗（RT）。当未使用Mohs或其他形式的PDEMA时，若主治医师认为切缘较窄，则可考虑采用放射治疗。若未预先进行RT治疗，则无法实施进一步切除术。

[继续](#)



Updates in Version 1.2026 of the NCCN Guidelines for Dermatofibrosarcoma Protuberans from Version 1.2025 include:

[DFSP-3](#)

- New header added: Follow-Up.
 - ▶ New bullets added:
 - ◊ Physical exam with focus on primary site every 6–12 months.
 - ◊ Patient education about regular self-exam.
 - ◊ Consider MRI surveillance when clinical exam is insufficient to assess local recurrence.
- Footnotes revised:
 - ▶ Footnote h: When Mohs or other forms of PDEMA are utilized and margins are negative, RT is not recommended. When Mohs or other forms of PDEMA are not utilized, consider RT if margins are considered narrow by the treating physicians *and further resection would not be feasible upon recurrence*. ~~RT if not given previously and further resection is not feasible.~~
 - ▶ Footnote i: Confirm ~~tumor mutation with~~ *COL1A1::PDGFB translocation* by fluorescence in situ hybridization (FISH) ~~for the translocation of PDGRF~~. Navarrete-Dechent C, et al. JAMA Dermatol 2019;155:361-369.

[DFSP-A](#)

- Fifth bullet revised: For equivocal lesions, consider fluorescence in situ hybridization (FISH), polymerase chain reaction (PCR), or conventional cytogenetics to detect t(17;22)(q22;q13), which is a hallmark of DFSP. Fusion of the ~~collagen type I alpha 1 gene (COL1A1)~~ at 17q22, with the ~~platelet-derived growth factor Beta gene (PDGFB)~~ at 22q13, form the oncogenic chimeric fusion gene *COL1A1::PDGFB*.

[DFSP-C](#)

- General Treatment Information, Adjuvant RT, Negative Margins, second sub-bullet revised: When Mohs or other forms of PDEMA are not utilized, consider RT if margins are considered narrow by the treating physician *and further resection would not be feasible upon recurrence*, ~~RT not given previously, and further resection is not feasible.~~



NCCN皮肤纤维肉瘤指南（版本1.2025）升级至版本1.2026后的主要更新包括：

DFSP-3

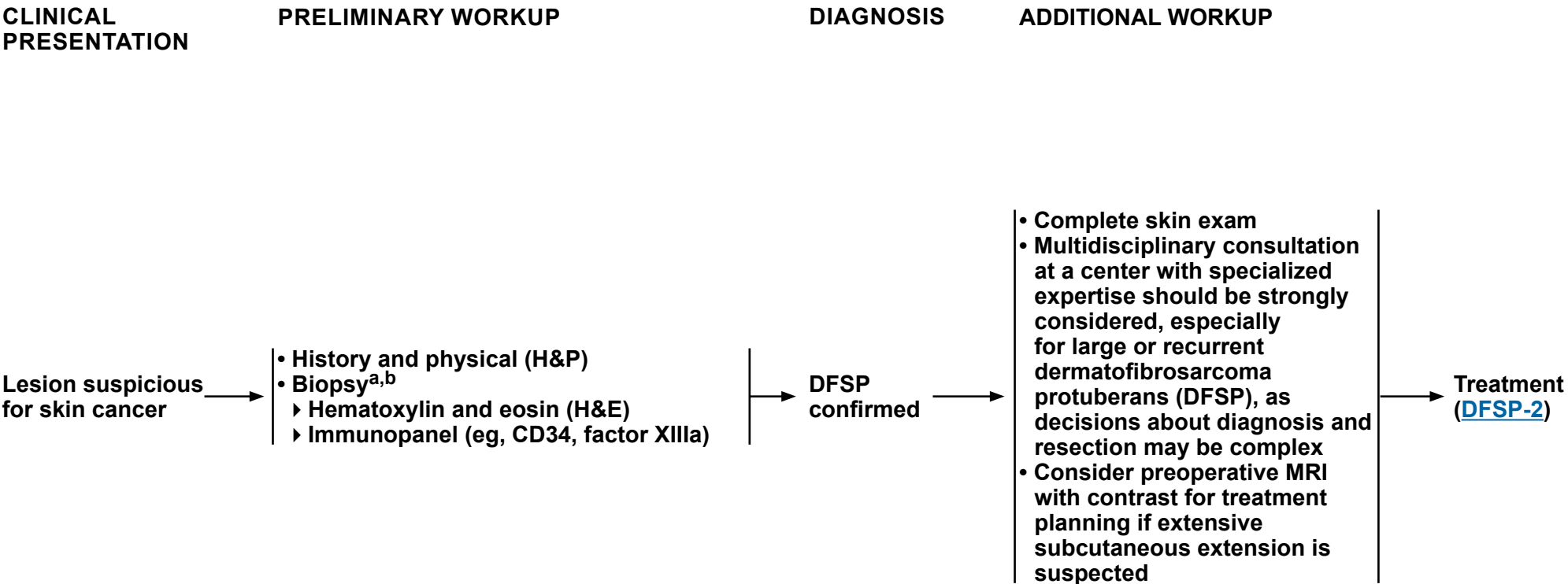
- 新增标题：后续跟进。
 - 新增了新子弹：
 - ◇ 每6至12个月进行一次以原发部位为重点的体格检查。
 - ◇ 关于定期自我检查的患者教育。
 - ◇ 当临床检查不足以评估局部复发时，应考虑进行MRI监测。
- 脚注已修订：
 - 脚注h：当使用Mohs或其他形式的PDEMA且边际值为负时，不建议采用RT方法。当未使用莫氏手术或其他形式的PDEMA时，若主治医师认为切缘较窄且复发后无法进行进一步切除，则可考虑行RT。若未预先进行RT且无法进一步切除。
 - 脚注i：通过荧光原位杂交（FISH）确认COL1A1::PDGFB易位是否导致肿瘤突变，以检测PDGRF易位。Navarrete-Dechent C等人。JAMA Dermatol 2019; 155:361-369。

DFSP-A

- 第五条修订内容：对于诊断不明确病变，建议采用荧光原位杂交（FISH）、聚合酶链反应（PCR）或常规细胞遗传学检测方法来识别t(17; 22)(q22; q13)易位——该易位是DFSP的典型特征。I型胶原 $\alpha 1$ 基因（COL1A1）位于17q22区域，与血小板衍生生长因子 β 基因（PDGFB β ）发生融合在22q13位点，形成了致癌性嵌合融合基因COL1A1::PDGFB β 。

DFSP-C

- 一般治疗信息、辅助放疗、阴性切缘、第二项子要点修订：当未采用莫氏手术或其他形式的PDEMA时，若主治医师认为切缘较窄且复发后无法进行进一步切除，则应考虑放疗。既往未行RT治疗，且无法进一步实施切除术。



^a This tumor is frequently misdiagnosed due to inadequate tissue sampling/superficial biopsy. Punch, incisional, or core biopsy, preferably including the deeper subcutaneous layer, is strongly recommended for sufficient tissue sampling and accurate pathologic assessment. If biopsy is indeterminate or clinical suspicion remains, rebiopsy is recommended. Wide undermining is discouraged due to the difficulty of interpreting subsequent re-excisions pathologically.

^b [Principles of Pathology \(DFSP-A\)](#).

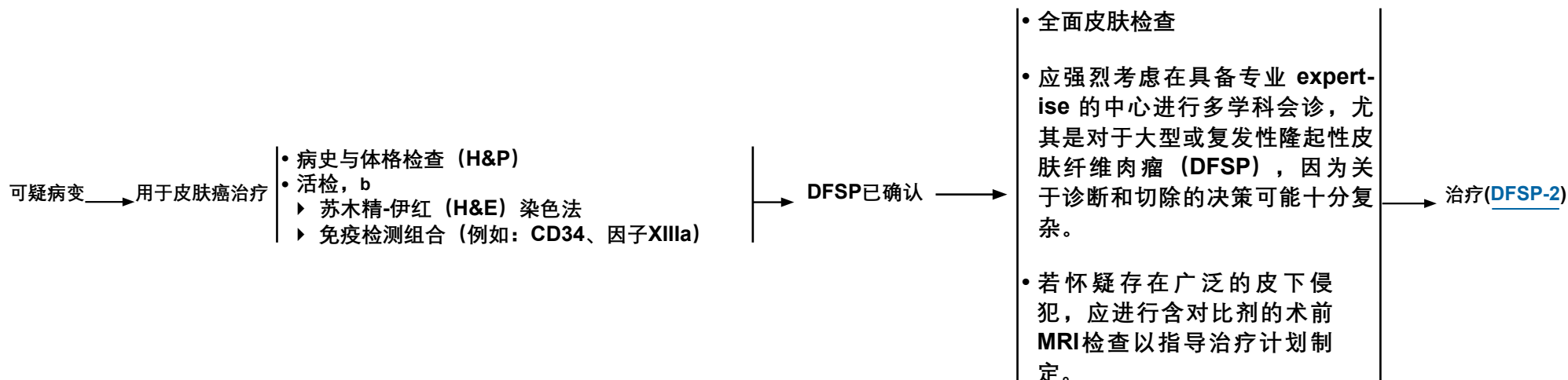
Note: All recommendations are category 2A unless otherwise indicated.

临床的
演讲；陈述

初步检查

诊断

进一步检查



^a 由于组织取样不足或活检深度不足，该肿瘤常被误诊。强烈建议采用穿刺活检、切开活检或核心活检（最好包括更深的皮下层），以确保获取足够的组织样本并进行准确的病理学评估。若活检结果无法确定或临床仍存疑虑，建议进行再次活检。由于后续再次切除标本的病理学判读存在困难，不推荐进行广泛浸润性病变切除术。

^b [病理学原理 \(DFSP-A\)](#)。

注：除非另有说明，所有建议均属于2A类。

TREATMENT

Excision with Mohs or other forms of peripheral and deep en face margin assessment (PDEMA)^{c,d,e}

- Debulking specimens from all excisions should be examined to identify fibrosarcomatous transformation of dermatofibrosarcoma protuberans (FS-DFSP)^c
- For unresectable/borderline resectable disease, consider tumor mutation analysis and neoadjuvant Imatinib^f

Negative surgical margins

Positive surgical margins

Re-resection^{d,e} until margins clear or surgery is not possible

Negative margins

Positive margins

ADJUVANT TREATMENT

Observation

Consider radiation therapy (RT)^g if margins are deemed narrow by treating physician and further resection would not be feasible upon recurrence

Follow-up ([DFSP-3](#))

^c If areas of transformation to fibrosarcoma or other sarcoma subtypes are identified, multidisciplinary consultation for consideration of further treatment and surveillance is recommended. FS-DFSP is associated with a higher metastasis risk. The patient should be referred to a center with expertise in management of soft tissue sarcomas. See [NCCN Guidelines for Soft Tissue Sarcoma](#) for multimodal therapy and surveillance considerations including CT of chest.

^d The most commonly used form of PDEMA is Mohs. See [NCCN Guidelines for Squamous Cell Skin Cancer - Principles of PDEMA Technique](#). When anatomic structures at the deep margin (eg, major vessels, nerves, bone) preclude complete histologic evaluation of the marginal surface, Mohs or other forms of PDEMA should be used to evaluate as much of the marginal surface as feasible. Treatment considerations for non-visualized areas may be the subject of multidisciplinary discussion.

^e If PDEMA is unavailable, consider wide excision. Wide undermining is discouraged prior to confirmation of clear margins due to the difficulty of interpreting subsequent re-excised margins, and the risk of concealing residual tumor below mobilized tissue. See [Principles of Surgery \(DFSP-B\)](#). Curtis KK, et al. J Natl Compr Canc Netw 2024;22:e247036.

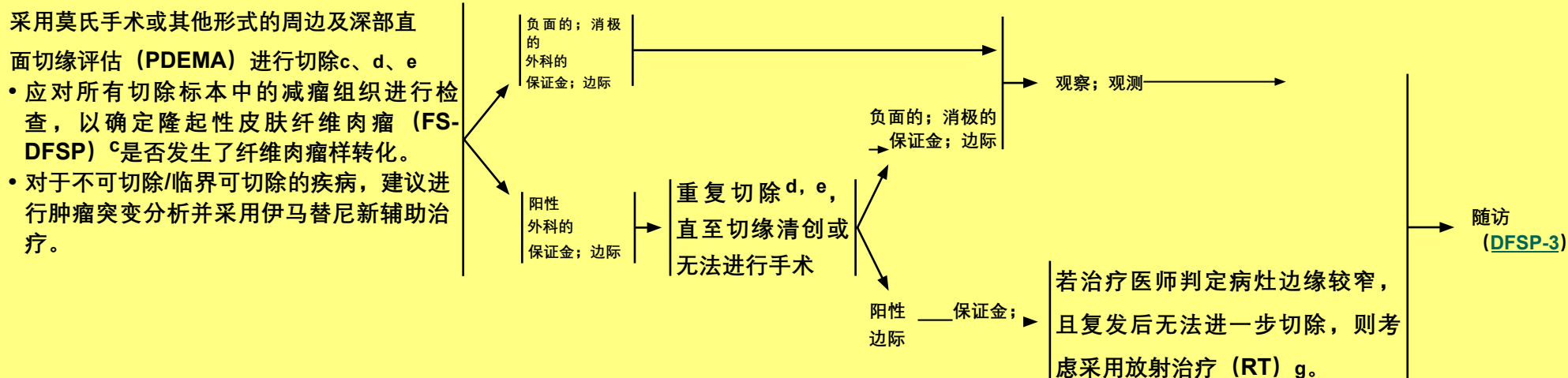
^f Consider neoadjuvant Imatinib for patients in whom resection with negative margins may result in unacceptable functional or cosmetic outcomes. Ugurel S, et al. Clin Cancer Res 2014;20:499-510.

^g [Principles of Radiation Therapy \(DFSP-C\)](#).

Note: All recommendations are category 2A unless otherwise indicated.

治疗

佐剂
治疗



^c 若发现病变区域向纤维肉瘤或其他肉瘤亚型转化，建议进行多学科会诊以评估进一步治疗方案及监测措施。FS-DFSP患者具有更高的转移风险。该患者应转诊至擅长软组织肉瘤治疗的中心。关于多模式治疗及监测措施（包括胸部CT检查），请参阅[《NCCN软组织肉瘤指南》](#)。

^d PDEMA最常用的形式是Mohs法。详见[《NCCN鳞状细胞皮肤癌指南——PDEMA技术原则》](#)。当深部边缘的解剖结构（如主要血管、神经、骨骼）阻碍对边缘表面进行完整的组织学评估时，应采用莫氏切片或其他形式的PDEMA技术，在可行范围内尽可能全面地评估边缘表面。对于无法显像区域的治疗方案需经多学科会讨论。若PDEMA不可用，则应考虑行广泛切除术。在确认切缘清晰之前，不建议进行广泛切缘扩大，因为后续重新切除的切缘难以准确判读，且存在使游离组织下方残留肿瘤被掩盖的风险。详见[《外科学原理 \(DFSP-B\)》](#)。Curtis KK 等。《美国国家综合癌症网络杂志》2024；22:e247036。

^f 对于那些经切缘阴性切除术后仍可能出现不可接受的功能或美容效果的患者，可考虑采用伊马替尼进行新辅助治疗。Ugurel S 等人，《临床癌症研究》2014；20：499-510。

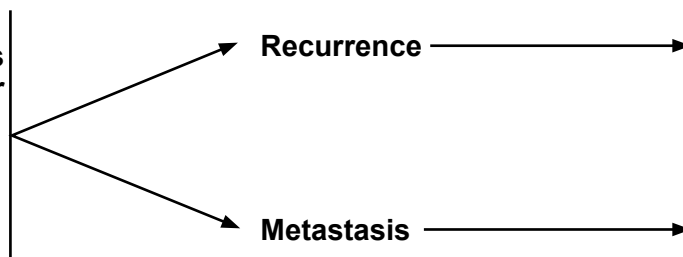
^g [放射治疗原理 \(DFSP-C\)](#)。

注：除非另有说明，所有建议均属于2A类。



FOLLOW-UP

- Physical exam with focus on primary site every 6–12 months
- Patient education about regular self-exam
- Consider MRI surveillance when clinical exam is insufficient to assess local recurrence



THERAPY FOR RECURRENCE/METASTASIS

Re-resection if feasible (preferred)^{d,e}
or
RT^{g,h} if not given previously and resection is not feasible
or
Consider Imatinibⁱ in cases where disease is unresectable,
or unacceptable functional or cosmetic outcomes will occur with resection

Multidisciplinary consultation^j

^d The most commonly used form of PDEMA is Mohs. See [NCCN Guidelines for Squamous Cell Skin Cancer - Principles of PDEMA Technique](#). When anatomic structures at the deep margin (eg, major vessels, nerves, bone) preclude complete histologic evaluation of the marginal surface, Mohs or other forms of PDEMA should be used to evaluate as much of the marginal surface as feasible. Treatment considerations for non-visualized areas may be the subject of multidisciplinary discussion.

^e If PDEMA is unavailable, consider wide excision. Wide undermining is discouraged prior to confirmation of clear margins due to the difficulty of interpreting subsequent re-excised margins, and the risk of concealing residual tumor below mobilized tissue. See [Principles of Surgery \(DFSP-B\)](#). Curtis KK, et al. J Natl Compr Canc Netw 2024;22:e247036.

^g [Principles of Radiation Therapy \(DFSP-C\)](#).

^h When Mohs or other forms of PDEMA are utilized and margins are negative, RT is not recommended. When Mohs or other forms of PDEMA are not utilized, consider RT if margins are considered narrow by the treating physicians and further resection would not be feasible upon recurrence.

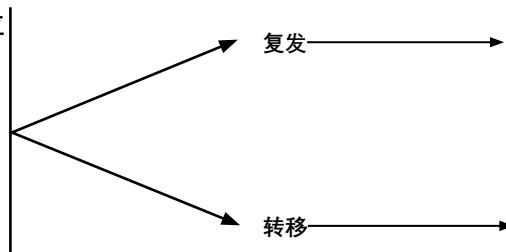
ⁱ Confirm *COL1A1::PDGFB* translocation by fluorescence in situ hybridization (FISH). Navarrete-Dechent C, et al. JAMA Dermatol 2019;155:361-369.

^j [NCCN Guidelines for Soft Tissue Sarcoma \(Synchronous STAGE IV \[EXTSARC-5\]\)](#).

Note: All recommendations are category 2A unless otherwise indicated.

随访

- 每6至12个月进行一次以原发部位为重点的体格检查
- 关于定期自我检查的患者教育
- 当临床检查不足以评估局部复发时，应考虑进行MRI监测。



复发/转移的治疗

如可行（首选）可再次切除^{d, e}
或者
RT^g, ^h若此前未给予治疗且无法进行切除术或在疾病无法切除，或切除术后会导致不可接受的功能或美容效果时，应考虑使用伊马替尼。

多学科会诊

^d PDEMA最常用的形式是Mohs法。详见《[NCCN鳞状细胞皮肤癌指南——PDEMA技术原则](#)》。当深部边缘的解剖结构（如主要血管、神经、骨骼）阻碍对边缘表面进行完整的组织学评估时，应采用莫氏切片或其他形式的PDEMA技术，在可行范围内尽可能全面地评估边缘表面。对于无法显像区域的治疗方案需经多学科会讨论确定。

^e 若无法获取PDEMA样本，应考虑行广泛切除术。在确认切缘阴性之前，不建议进行广泛底切术，因为这会增加后续再次切除切缘的判读难度，并存在将残留肿瘤隐藏在游离组织下方的风险。详见《[外科学原理\(DFSP-B\)](#)》。Curtis KK 等人，《美国国家综合癌症网络杂志》2024；22:e247036。

^g [放射治疗原理 \(DFSP-C\)](#)。

^h 当使用莫氏法或其他形式的PDEMA且切缘呈阴性时，不推荐采用放射治疗（RT）。当未使用莫氏手术或其他形式的PDEMA时，若主治医师认为切缘过窄且复发后无法进行进一步切除，则应考虑行RT（放射治疗）。

ⁱ 通过荧光原位杂交（FISH）确认COL1A1::PDGFB易位。Navarrete-Dechent C等。《JAMA Dermatol》2019；155:361-369。

^j [NCCN软组织肉瘤指南（同步分期IV期\[EXTSARC-5\]）](#)。

注：除非另有说明，所有建议均属于2A类。



PRINCIPLES OF PATHOLOGY^a

- Evaluation by a qualified physician with specific expertise in sarcoma/soft tissue pathology or dermatopathology is preferred (if available).
- The spindled cells arranged in a storiform or fascicular pattern are typically bland with minimal cytologic atypia.
- Immunohistochemistry (IHC) for CD34 is mostly positive, and factor XIIIa negative.
- FS-DFSP^b is characterized by transition from storiform to a herringbone pattern, with a higher degree of cellularity, cytologic atypia, mitotic activity (>5/10 high-power fields [HPFs]), and frequent loss of CD34 immunostaining. When CD34 is negative, other markers such as S100 should also be negative to rule out other spindle cell tumors.
- For equivocal lesions, consider fluorescence in situ hybridization (FISH), polymerase chain reaction (PCR), or conventional cytogenetics to detect t(17;22)(q22;q13), which is a hallmark of DFSP. Fusion of *COL1A1* at 17q22, with *PDGFB* at 22q13, form the oncogenic chimeric fusion gene *COL1A1::PDGFB*.
- Margin control during excision (see [Principles of Surgery \[DFSP-B\]](#)) may occasionally be aided by H&E sections supplemented by CD34 IHC.

^a Currently, no American Joint Committee on Cancer (AJCC) or College of American Pathologists (CAP) synoptic reporting is defined.

^b If areas of transformation to fibrosarcoma or other sarcoma subtypes are identified, multidisciplinary consultation for consideration of further treatment and surveillance is recommended. FS-DFSP is associated with a higher metastasis risk. The patient should be referred to a center with expertise in management of soft tissue sarcomas. See [NCCN Guidelines for Soft Tissue Sarcoma](#) for multimodal therapy and surveillance considerations including CT of draining nodal basin and chest.

Note: All recommendations are category 2A unless otherwise indicated.

病理学原理^a

- 建议由具备肉瘤/软组织病理学或皮肤病理学特定专业知识的合格医师进行评估（如条件允许）。
- 呈网状或束状排列的梭形细胞通常形态平滑，细胞异型性极轻微。
- CD34的免疫组织化学（IHC）检测结果大多呈阳性，而XIIIa因子检测结果为阴性。
- FS-DFSPb的特征是从栅格状结构转变为人字形模式，同时具有更高的细胞密度、细胞学非典型性、有丝分裂活性（>5/10高倍视野[HPF]），以及CD34免疫染色频繁缺失。当CD34呈阴性时，其他标志物（如S100）也应呈阴性，以排除其他梭形细胞肿瘤。
- 对于诊断不明确病变，可考虑采用荧光原位杂交（FISH）、聚合酶链反应（PCR）或常规细胞遗传学检查以检测t(17; 22)(q22; q13)易位——该易位是DFSP的典型特征。位于17q22的COL1A1与位于22q13的PDGFB融合，形成了致癌性嵌合融合基因COL1A1::PDGFB。
- 切除过程中的切缘控制（参见[《外科学原理》\[DFSP-B\]](#)）有时可通过苏木精-伊红（H&E）切片结合CD34免疫组化（IHC）检测获得辅助。

^a 目前，美国癌症联合委员会（AJCC）或美国病理学家学会（CAP）尚未制定统一的综合报告标准。

^b 若发现病变区域向纤维肉瘤或其他肉瘤亚型转化，建议进行多学科会诊以评估进一步治疗方案及监测措施。FS-DFSP患者具有更高的转移风险。该患者应转诊至擅长软组织肉瘤治疗的专科中心。关于多模式治疗及监测方案（包括引流淋巴结区域及胸部CT检查），请参阅[《NCCN软组织肉瘤指南》](#)。

注：除非另有说明，所有建议均属于2A类。



PRINCIPLES OF SURGERY

Goal:

- Every effort should be made to achieve clear surgical margins. Complete histologic surgical margin examination to include the entire excised peripheral and deep margin is recommended, whenever possible. Tumor characteristics include long, irregular, subclinical extensions. Debulking specimens from all excisions should be examined to identify FS-DFSP since this is associated with metastatic potential.

Surgical Approach: Mohs or Other Forms of PDEMA

- [NCCN Guidelines for Squamous Cell Skin Cancer - Principles of PDEMA Technique](#)
- If Mohs or other forms of PDEMA are unavailable, consider wide excision.¹
 - ▶ Reconstruction should be delayed until clear margins have been verified to avoid the risk of translocating the tumor within the resection bed, thus making further margin assessments inaccurate.

¹ Farma JM, Ammori JB, Zager JS, et al. Dermatofibrosarcoma protuberans: how wide should we resect? Ann Surg Oncol 2010;17:2112-2118.

Note: All recommendations are category 2A unless otherwise indicated.

外科手术原理

目标：

- 应尽一切努力实现清晰的手术切缘。建议在可能的情况下，对切除的周边及深部切缘进行全面的组织学检查。肿瘤特征包括长而不规则的亚临床延伸。所有切除标本均应进行减瘤病理检查以识别**FS-DFSP**，因为该病变与转移潜能相关。

手术方法：莫氏手术或其他形式的**PDEMA**• [NCCN鳞状细胞皮肤癌指南——PDEMA技术原则](#)

- 若无法使用莫氏制剂或其他形式的**PDEMA**，应考虑行广泛切除术。¹
 - ▶ ~~应延迟重建手术，直至确认切缘清晰，以避免肿瘤在切除床上移位~~的风险，从而确保后续切缘评估的准确性。

¹ Farma JM, Ammori JB, Zager JS, 等.隆突性皮肤纤维肉瘤:应切除多广泛?《外科肿瘤学年鉴》2010;17:2112-2118.

注：除非另有说明，所有建议均属于**2A**类。



PRINCIPLES OF RADIATION THERAPY

General Treatment Information

• Adjuvant RT:

▶ Positive Margins/Gross Disease

- ◊ 50–60 Gy for indeterminate or positive margins, and up to 66 Gy for positive margins or gross tumor (2-Gy fractions per day).
- ◊ Fields to extend widely beyond surgical margins (eg, 3–5 cm) when clinically feasible.

▶ Negative Margins

- ◊ When Mohs or other forms of PDEMA are utilized and margins are negative, RT is not recommended.
- ◊ When Mohs or other forms of PDEMA are not utilized, consider RT if margins are considered narrow by the treating physician and further resection would not be feasible upon recurrence.

• Recurrence/Metastasis:

- ▶ RT if not given previously and further resection is not feasible; 50–60 Gy for indeterminate or positive margins, and up to 66 Gy for positive margins or gross tumor (2-Gy fractions per day).
- ▶ Fields to extend widely beyond surgical margins (eg, 3–5 cm) when clinically feasible.

Note: All recommendations are category 2A unless otherwise indicated.

放射治疗原理

一般治疗信息

• 辅助放疗：

▶ 阳性边缘值/总疾病量

◇ 对于切缘不确定或阳性区域，照射剂量为**50–60 Gy**；对于阳性切缘或大体肿瘤区域，照射剂量最高可达**66 Gy**（每日分次照射剂量为**2 Gy**）。

◇ 在临床可行的情况下，手术切缘范围应广泛延伸至远超手术边缘（例如**3–5厘米**）。

▶ 负利润率

◇ 当使用莫氏法或其他形式的**PDEMA**且切缘呈阴性时，不推荐采用放射治疗（RT）。

◇ 当未使用莫氏手术或其他形式的**PDEMA**时，若主治医师认为切缘过窄且复发后无法进行进一步切除，则应考虑行RT（放射治疗）。

• 复发/转移：

▶ 若未预先进行放疗且无法进一步切除；对于切缘不确定或阳性区域，放疗剂量为**50–60 Gy**；对于切缘阳性或肉眼可见肿瘤，放疗剂量最高可达**66 Gy**（每日分次照射**2 Gy**）。

▶ 在临床可行的情况下，手术切缘范围应广泛延伸至远超手术边缘（例如**3–5厘米**）。

注：除非另有说明，所有建议均属于**2A类**。



ABBREVIATIONS

CAP	College of American Pathologists
DFSP	dermatofibrosarcoma protuberans
FISH	fluorescence in situ hybridization
FS-DFSP	fibrosarcomatous transformation of DFSP
H&E	hematoxylin and eosin
H&P	history and physical
HPF	high-power field
IHC	immunohistochemistry
PCR	polymerase chain reaction
PDEMA	peripheral and deep en face margin assessment

缩写

CAP	美国病理学家学会
DFSP	隆起性皮肤纤维肉瘤
鱼类；鱼类生物	荧光原位杂交
FS-DFSP	DFSP的纤维肉瘤样转化
H&E染色	苏木精和伊红
H&P	历史与物理
HPF	高功率场
免疫组化 (IHC)	免疫组织化学
聚合酶链式反应 (PCR)	聚合酶链式反应
PDEMA	周边及深层正面边缘评估



NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN Categories of Preference	
Preferred	Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended	Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances	Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

NCCN证据等级与共识		
类别	1	根据高级别证据（≥1项随机3期临床试验或高质量、稳健的荟萃分析），NCCN共识一致认为该干预措施是适宜的（专家组支持率≥85%）。
类别	2A	根据较低层级的证据，NCCN达成一致共识（专家组支持率≥85%），认为该干预措施是适宜的。
类别	2B	根据较低层级的证据，NCCN达成共识（专家组支持率≥50%但<85%），认为该干预措施是适宜的。
类别	3	基于任何证据等级，NCCN均对干预措施的适用性存在重大分歧。

除非另有说明，所有建议均属于2A类。

NCCN偏好分类	
首选的	基于卓越疗效、安全性及证据支持的干预措施；在适当情况下，还需具备可负担性。
其他推荐	其他干预措施可能效果稍逊、毒性更强，或基于尚不成熟的数据；或者在获得类似疗效的情况下成本显著更高。
在某些情况下非常有用。	其他可用于特定患者群体（根据推荐标准定义）的干预措施。

所有建议均被认为恰当。



Discussion

This discussion corresponds to the NCCN Guidelines for Dermatofibrosarcoma Protuberans. Last updated: October 24, 2025

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讨论

本讨论依据美国国家综合癌症网络（NCCN）发布的《隆起性皮肤纤维肉瘤诊疗指南》进行。最后更新日期：2025年10月24日。

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Overview

Dermatofibrosarcoma protuberans (DFSP) is an uncommon, low-grade sarcoma of fibroblast origin with an incidence rate of 4.1 to 4.5 cases per million persons per year in the United States.¹⁻⁴ A predilection for occurring in African Americans has been reported in one study.³ They are often clinically misdiagnosed, leading to prolonged time to diagnosis. However, DFSP rarely metastasizes.⁵ When metastasis occurs, it is typically in the lung, bone, or regional lymph nodes. Three-dimensional reconstruction of DFSP⁶ has revealed tumors with highly irregular shapes and frequent finger-like extensions.⁷ As a result, incomplete removal and subsequent recurrence are common without attention to full assessment of the peripheral and deep margin. The local recurrence rate for wide local excision (WLE) of DFSP in studies ranges from 10% to 60%, whereas the rate of development of regional or distant metastatic disease is only 1% and 4-7.4%, respectively.^{8,9}

Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Guidelines are available at www.NCCN.org.

Literature Search Criteria and Guidelines Update Methodology

Prior to the update of this version of the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Dermatofibrosarcoma Protuberans, an electronic search of the PubMed database was performed to obtain key literature using the following search term: dermatofibrosarcoma protuberans. The PubMed database was chosen as it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.¹⁰

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article

types: Clinical Trial; Guideline; Meta-Analysis; Practice Guideline; Randomized Controlled Trial; Systematic Reviews; and Validation Studies.

The data from key PubMed articles as well as articles from additional sources deemed as relevant to these guidelines as discussed by the Panel during the Guidelines update have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

Sensitive/Inclusive Language

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation.¹¹ NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms men, women, female, and male when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

概述

隆突性皮肤纤维肉瘤（DFSP）是一种罕见的低度恶性纤维母细胞来源性肉瘤，在美国的年发病率为每百万人4.1至4.5例。¹⁻⁴一项研究报道指出，该疾病在非裔美国人中更为常见。³这类患者常被临床误诊，导致诊断延迟。然而，DFSP极少发生转移。⁵当发生转移时，通常发生在肺部、骨骼或区域淋巴结。

对DFSP⁶的三维重建显示，肿瘤形态高度不规则，且常伴有指状延伸结构。⁷因此，若未对周围及深部切缘进行全面评估，肿瘤切除不彻底及后续复发的情况十分常见。多项研究显示，DFSP采用广泛局部切除术（WLE）后的局部复发率介于10%至60%之间，而区域转移或远处转移的发生率分别仅为1%和4%-7.4%。^{8,9}

指南更新方法论

NCCN指南的制定与更新完整细节可访问www.NCCN.org获取。

文献检索标准与指南的更新方法论

在本版NCCN肿瘤学临床实践指南（NCCN指南®）更新之前针对隆起性皮肤纤维肉瘤，我们对PubMed数据库进行了电子检索，使用以下检索词获取关键文献：

隆起性皮肤纤维肉瘤。选择PubMed数据库是因为它仍是医学文献最广泛使用的资源，并收录了同行评审的生物医学文献。¹⁰

通过筛选以英文发表的人体研究，最终缩小了搜索结果范围。研究结果仅限于以下文章类型：临床试验；指南；荟萃分析；实践指

南；随机对照试验；系统评价；以及验证研究。

本讨论部分收录了来自关键PubMed文章的数据，以及专家组在指南更新过程中认为与这些指南相关的其他来源文章的数据。对于缺乏高级别证据的建议，是基于专家组对较低级别证据及专家意见的评估得出的。

敏感/包容性语言

NCCN指南力求采用能够促进公平性、包容性及代表性目标的语言表述。¹¹ NCCN指南力求采用以患者为中心的语言表述，避免污名化，并反对种族主义、阶级歧视、厌女主义、年龄歧视、能力歧视及体重偏见。该指南涵盖所有性取向和性别认同的个体。NCCN指南采用非性别化表述，转而侧重于器官特异性建议。这种语言不仅更为准确，也更具包容性，能够全面满足所有性取向和性别认同个体的需求。NCCN指南在引用未采用包容性术语的机构或来源提供的统计数据、建议或数据时，将继续使用“男性”、“女性”、“女”及“男”等术语。大多数研究并未详细说明性别与性别认同数据的收集方式，且这些术语在研究中被交替使用或使用不一致。若资料来源未区分出生时性别与出生时指定的性别或现存器官，该信息应被推定主要代表顺性别个体。NCCN鼓励研究人员在未来的研究和组织工作中收集更具体的数据。在未来的分析中采用更具包容性且更准确的语言。



Evaluation

Preliminary workup for DFSP consists of history and physical (H&P) examination, and biopsy. It should be noted that this tumor is frequently misdiagnosed due to inadequate tissue sampling resulting from shallow biopsy. As the superficial aspect of a DFSP may not be distinct from benign lesions, a punch or incisional biopsy that samples the subcutaneous layer is strongly recommended. If a biopsy is indeterminate or clinical suspicion remains, rebiopsy is recommended. Histologically, DFSP typically presents as a storiform or fascicular proliferation of bland spindled cells that extends from the dermis into the subcutis.^{12,13} Virtually all cases are immunohistologically CD34-positive and factor XIIIa-negative with rare exceptions.^{14,15} Currently, no synoptic reporting is recommended.

In most cases, examination of hematoxylin and eosin-stained specimens by light microscopy results in an unequivocal diagnosis. However, differentiation of DFSP from dermatofibroma can be difficult at times. Staining with CD34,^{15,16} factor XIIIa,^{14,17} and other immunomarkers such as stromelysin 3, nestin, apolipoprotein D, and cathepsin K,¹⁸⁻²¹ might be useful in such instances. The NCCN Panel recommends that appropriate and confirmatory immunostaining be performed in all cases of suspected DFSP.

After DFSP confirmation, additional workup may include a complete skin examination as well as consideration of preoperative MRI with contrast for treatment planning if there is suspicion of extensive subcutaneous extension. As decisions about diagnosis and resection may be complex, multidisciplinary consultation at a center with specialized expertise should be strongly considered, especially for large or recurrent DFSP as it may optimize clinical and reconstructive outcomes.^{22,23}

Treatment

Initial treatment of DFSP is surgical. Because of its proclivity for irregular and frequently deep subclinical extensions, every effort should be made to remove this tumor in its entirety at the time of initial therapy. Excision with Mohs micrographic surgery (Mohs) or other forms of peripheral and deep en face margin assessment (PDEMA) is recommended over WLE. En face sectioning is preferred to prevent missing small foci of tumor. The most commonly used form of PDEMA is Mohs (See NCCN Guidelines for Squamous Cell Skin Cancer – Principles of PDEMA Technique, available at www.NCCN.org).²⁴ When anatomic structures at the deep margin (eg, major vessels, nerves, bone) preclude complete histologic evaluation of the marginal surface via Mohs or other forms of PDEMA, these surgical techniques should be used to evaluate as much of the marginal surface as feasible. A combination of PDEMA and WLE for the deep margin has been reported in the literature.²³ Treatment considerations for non-visualized areas may be the subject of multidisciplinary discussion. If PDEMA is unavailable, WLE can be considered. Wide undermining is discouraged prior to confirmation of clear margins due to the difficulty of interpreting subsequent re-excised margins, and the risk of concealing residual tumor below mobilized tissue.

It is unclear whether the histologic features of a high mitotic rate or evidence of fibrosarcomatous (FS)-DFSP change have prognostic significance in DFSP.^{25,26} Studies in the biomedical literature both support²⁷⁻³⁵ the connection between FS-DFSP and an increased risk of local recurrence, lower time to recurrence, and increased risk of metastasis, and refute^{36,37} this notion. A systematic review of 225 patients with FS-DFSP and 1422 with DFSP reported risks of local recurrence (29.8% vs. 13.7%), metastasis (14.4% vs. 1.1%), and death (14.7% vs. 0.8%) from the disease to be significantly higher in FS-DFSP.³⁸ Overall, FS-DFSP is associated with a metastatic risk range of

评估

DFSP的初步检查包括病史采集、体格检查（H&P）及活检。需要注意的是，由于浅表活检导致组织取样不足，该肿瘤常被误诊。由于DFSP的表面表现可能与良性病变难以区分，强烈建议采用穿刺或切开活检术取皮下组织样本。若活检结果无法明确诊断或临床仍存疑虑，则建议进行再次活检。组织学上，DFSP通常表现为平滑梭形细胞的栅状或束状增生，从真皮层延伸至皮下组织。^{12,13}几乎所有病例在免疫组织化学检测中均呈CD34阳性、XIIIa因子阴性，仅存在极少数例外。^{14,15}目前不推荐进行综合报告。

在大多数情况下，通过光学显微镜检查苏木精-伊红染色标本可得出明确诊断。然而，有时难以区分DFSP与皮肤纤维瘤。在此类情况下，采用CD34^{15,16}、XIIIa因子^{14,17}以及其他免疫标志物（如基质溶素3、巢蛋白、载脂蛋白D和组织蛋白酶K¹⁸⁻²¹）进行染色可能具有实用价值。NCCN专家组建议，在所有疑似DFSP病例中均应进行恰当且具有确认性的免疫组化染色检查。

经DFSP确认后，进一步检查可能包括全面的皮肤检查；若怀疑肿瘤存在广泛的皮下侵犯，则需考虑进行术前增强MRI以指导治疗方案制定。由于关于诊断和切除的决策可能较为复杂，应强烈考虑在具备专业 expertise 的中心进行多学科会诊。尤其对于大型或复发性DFSP，因为这可能优化临床及重建效果。^{22、23}

治疗

DFSP的初始治疗方式为手术切除。由于该肿瘤易出现不规则且常为深部的亚临床侵犯，应在初始治疗时尽一切努力实现肿瘤的完全切除。推荐采用莫氏显微外科手术（Mohs）或其他形式的周边及深部直面切缘评估（PDEMA）进行切除，而非全层切除术（WLE）。首选直面切片技术可避免遗漏微小肿瘤病灶。最常用的PDEMA形式是Mohs法（详见NCCN《鳞状细胞皮肤癌指南》——PDEMA技术原则，网址：www.NCCN.org）。²⁴当深部边缘的解剖结构（例如主要血管、神经、骨骼）阻碍了通过莫氏法或其他形式的PDEMA对边缘表面进行完整的组织学评估时，应采用这些手术技术尽可能全面地评估边缘表面。文献报道可联合使用PDEMA与WLE技术处理深部边缘。²³对于无法显像区域的治疗方案可由多学科团队共同讨论。若无法采用PDEMA技术，可考虑采用WLE技术。不建议进行广泛的骨下剥离术。在确认切缘清晰之前，需考虑以下因素：后续再次切除切缘的解读存在困难，且存在将残留肿瘤隐藏在游离组织下方的风险。

目前尚不清楚高有丝分裂率的组织学特征或纤维肉瘤样（FS）-DFSP改变是否对DFSP具有预后意义。^{25,26}生物医学文献中的研究均支持²⁷⁻³⁵ FS-DFSP与局部复发风险增加、复发时间缩短以及转移风险升高之间的关联，并驳斥了^{36,37}关于这一观点，一项针对225例FS-DFSP患者和1422例DFSP患者的系统性综述表明，FS-DFSP患者的局部复发风险（29.8% 对 13.7%）、转移风险（14.4% 对 1.1%）及死亡风险（14.7% 对 0.8%）均显著更高。³⁸总体而言，FS-DFSP与转移风险范围相关。



10% to 23.5%.^{26,29,39} The NCCN Panel recommends that the debulking specimens from all excisions should be examined to identify FS transformation of DFSP. If FS transformation or other sarcoma subtypes are found, multidisciplinary consultation for consideration of further treatment and surveillance is recommended. Clinicians should consult the NCCN Guidelines for Soft Tissue Sarcoma (available at www.NCCN.org) for multimodal therapy and surveillance considerations including chest CT.

Additionally, tumor mutation analysis and neoadjuvant imatinib can be considered options for unresectable or borderline resectable disease. Consider neoadjuvant imatinib for patients in whom DFSP resection with negative margins may result in unacceptable functional or cosmetic outcomes.⁴⁰ The NCCN Panel recommends that if a negative margin is achieved, no adjuvant treatment is necessary. When Mohs or other forms of PDEMA are used, radiation therapy (RT) is not recommended.

If initial surgery yields positive margins, re-resection is recommended whenever possible, with the goal of achieving clear margins. Mohs or modified Mohs surgery, and traditional WLE with wider margins, which has been associated with higher tumor clearance and lower rates of recurrence,⁴¹⁻⁴³ are all methods to achieve complete histologic assessment. Studies examining outcomes of both Mohs and WLE have consistently reported lower recurrence rates for the former (0%–6.6% vs. 1.7%–30.8%).⁴⁴⁻⁵³ A large retrospective series of 204 patients with DFSP showed a very low local recurrence rate (1%) using WLE with total peripheral margin pathologic evaluation, underscoring the importance of meticulous pathologic margin evaluation with any surgical technique.⁵⁴ This notion was also supported by more studies.^{55,56} It is recommended that any reconstruction involving extensive undermining be avoided. Tissue rearrangement, if necessary, should be delayed until negative histologic margins are verified to prevent displacing a

potentially positive margin or hampering interpretation of re-excisions. If there is concern that the surgical margins are not clear when Mohs or PDEMA is not available, split-thickness skin grafting should be considered to monitor for recurrence.

While radiation has occasionally been used as a primary therapeutic modality for DFSP,⁵⁷⁻⁵⁹ it is most beneficial as adjuvant therapy after surgery.⁵⁷⁻⁶⁴ In a single-institution retrospective review of 53 patients with DFSP treated with surgery and preoperative or postoperative RT, local control was 93% and actuarial overall survival was 98% at 10 years.³⁷ Another small patient series reported that 86% of patients with DFSP who were treated with postoperative RT remained disease-free at a median follow-up of 10.5 years.⁶³ In a systematic review and meta-analysis of adjuvant RT for DFSP after WLE, the overall recurrence rate was reported to be 11.74%. Patients with positive/close margins had a recurrence rate of 14.23% whereas those with negative margins had no recurrence.⁶⁵ The NCCN Panel recommends that if margins are positive after re-resection, RT can be considered as an adjuvant treatment if the treating physician deems the margins to be narrow and further resection would not be feasible upon recurrence.

DFSP can be treated by targeted platelet-derived growth factor receptor (PDGFRs). DFSP is characterized by a translocation between chromosomes 17 and 22 [t(17;22)(q22;q13)] resulting in the overexpression of *PDGFB*.⁶⁶⁻⁶⁸ These findings suggest that targeting PDGFRs may be an effective treatment for DFSP. In published results, imatinib mesylate, a protein tyrosine kinase inhibitor, has shown clinical activity against DFSP,^{40,69-73} which has led to its approval by the U.S. Food and Drug Administration (FDA) for the treatment of unresectable, recurrent, and/or metastatic DFSP in adult patients. It is still unclear whether or not and the extent to which the *COL1A1::PDGFB* fusion gene dictates imatinib response.⁷⁰ In a systematic review that included

10%至23.5%。^{26,29,39} NCCN专家组建议应对所有切除术获得的减瘤标本进行检查，以确定DFSP是否发生FS转化。若发现FS转化或其他肉瘤亚型，建议进行多学科会诊以评估进一步治疗方案及监测措施。临床医生应参考《NCCN软组织肉瘤指南》（详见www.NCCN.org）中的多模式治疗及监测建议，包括胸部CT检查。

此外，对于不可切除或临界可切除的肿瘤，可考虑采用肿瘤突变分析及新辅助伊马替尼治疗方案。对于经DFSP切除术且切缘阴性后仍可能出现不可接受的功能或美容效果的患者，可考虑采用新辅助伊马替尼治疗。⁴⁰ NCCN专家组建议，若达到阴性切缘，则无需进行辅助治疗。当采用莫氏手术或其他形式的PDEMA时，不推荐进行放射治疗（RT）。

若初次手术切缘呈阳性，则建议在条件允许时再次切除，以实现切缘阴性。Mohs手术或改良Mohs手术，以及采用更广泛切缘的传统全层切除术（WLE）——后者已被证实能实现更高的肿瘤清除率和更低的复发率⁴¹⁻⁴³——均为实现完全组织学评估的有效方法。多项针对Mohs手术与全层切除术（WLE）疗效的研究一致表明，前者复发率更低（0%–6.6% 对比 1.7%–30.8%）。⁴⁴⁻⁵³ 一项针对204例DFSP患者的大型回顾性研究显示，采用全淋巴结清扫术（WLE）并结合外周切缘病理评估后，局部复发率极低（1%）。强调在任何手术技术中进行细致的病理切缘评估的重要性。⁵⁴ 这一观点也得到了更多研究的支持。^{55,56} 建议避免任何涉及广泛骨基质破坏的重建术式。如需进行组织重塑，应延迟至确认组织学切缘阴性后再实施，以防止移位。

手术切缘可能呈阳性或影响对再次切除术结果的解读。若担心在无法进行莫氏手术或PDEMA术式时手术切缘不清晰，应考虑采用中厚皮片移植术以监测复发情况。

尽管放射治疗偶尔被用作DFSP的主要治疗方式⁵⁷⁻⁵⁹，但其最有效的应用场景是作为术后辅助治疗。⁵⁷⁻⁶⁴ 在一项针对53例接受手术联合术前或术后放疗的DFSP患者的单中心回顾性研究中，局部控制率为93%，10年精算总生存率为98%。³⁷ 另一项小型患者系列研究报告显示，接受术后放疗的DFSP患者中，86%在中位随访10.5年时仍保持无病状态。⁶³ 在一项关于全肺切除术后DFSP辅助放疗的系统评价与荟萃分析中，总体复发率为11.74%。切缘阳性/闭合的患者复发率为14%。23%，而切缘阴性的患者未出现复发。⁶⁵ NCCN专家组建议：若再次切除后的切缘呈阳性，且主治医师认为切缘狭窄、无法进行进一步切除时，可考虑将放疗（RT）作为辅助治疗方案。复发。

DFSP可通过靶向血小板衍生生长因子受体（PDGFRs）进行治疗。DFSP的特征是17号和22号染色体之间的易位[t(17; 22)(q22; q13)]，导致PDGFB基因过度表达。⁶⁶⁻⁶⁸ 这些研究结果表明，靶向PDGFRs可能是治疗DFSP的有效方法。已发表的研究结果显示，蛋白酪氨酸激酶抑制剂甲磺酸伊马替尼对DFSP具有临床活性^{40,69-73}，因此获得了美国的批准。美国食品药品监督管理局（FDA）批准用于治疗成人患者的不可切除、复发性和/或转移性DFSP。目前尚不清楚COL1A1::PDGFB融合基因是否以及在何种程度上决定伊马替尼的治疗反应。⁷⁰ 一项纳入了多项研究的系统综述表明.....



patients receiving imatinib as monotherapy, adjuvant, or neoadjuvant therapy, complete response, partial response, stable disease, and progressive disease were reported in 5.2%, 55.2%, 27.6%, and 9.2% of patients, respectively.⁷⁰ In the neoadjuvant setting, complete response, partial response, stable disease, and progressive disease rates were reported to be 7.1%, 50%, 35.7%, and 7.1%, respectively.⁴⁰

Follow-up

Given the historically high local recurrence rates for DFSP, ongoing clinical follow-up with focus on the primary site every 6 to 12 months is indicated, with re-biopsy of any suspicious regions. Although metastatic disease is rare, a guided H&P and patient education about regular self-examination are recommended. Consider MRI surveillance for deeply invasive disease or other concerns related to recurrence.

Recurrent tumors should be resected whenever possible. Adjuvant RT may be considered after surgery. For patients who are not surgical candidates, RT alone is an option if not given previously. Imatinib mesylate should be considered in cases where the disease is unresectable following multiple resections, or if unacceptable functional or cosmetic outcomes will occur with further resection. The NCCN Panel recommends that the *COL1A1::PDGFB* translocation be confirmed by fluorescence in-situ hybridization (FISH).⁷⁰

In the rare event of metastatic disease, multidisciplinary consultation is recommended to coordinate treatment (See NCCN Guidelines for Soft Tissue Sarcoma, available at www.NCCN.org).

在接受伊马替尼作为单药治疗、辅助治疗或新辅助治疗的患者中，完全缓解、部分缓解、疾病稳定和疾病进展的发生率分别为5.2%、55.2%、27.6%和9.2%。⁷⁰在新辅助治疗方案中，完全缓解率、部分缓解率、疾病稳定率和疾病进展率分别为7.1%、50%、35.7%和7.1%。⁴⁰

随访

鉴于DFSP的局部复发率在历史上一直较高，建议进行持续临床随访，每6至12个月重点检查原发部位，并对任何可疑区域进行再次活检。尽管转移性疾病较为罕见，仍建议进行引导性直肠指检（H&P）并指导患者定期进行自我检查。对于深部浸润性疾病或其他与复发相关的疑虑，可考虑采用磁共振成像（MRI）进行监测。

复发性肿瘤应在条件允许时予以切除。术后可考虑辅助放疗（RT）。

对于不适合手术的患者，若此前未接受过放疗，单纯放疗亦可作为治疗选择。当疾病在多次切除术后仍无法切除，或进一步切除会导致不可接受的功能或外观后果时，应考虑使用甲磺酸伊马替尼。NCCN专家组建议采用荧光原位杂交（FISH）技术确认COL1A1::PDGFB易位。⁷⁰

在罕见的转移性疾病情况下，建议进行多学科会诊以协调治疗（详见NCCN软组织肉瘤指南，网址：www.NCCN.org）。



References

1. Criscione VD, Weinstock MA. Descriptive epidemiology of dermatofibrosarcoma protuberans in the United States, 1973 to 2002. *J Am Acad Dermatol* 2007;56:968–973. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/17141362>.
2. Rouhani P, Fletcher CD, Devesa SS, Toro JR. Cutaneous soft tissue sarcoma incidence patterns in the U.S. : an analysis of 12,114 cases. *Cancer* 2008;113:616–627. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/18618615>.
3. Kreicher KL, Kurlander DE, Gittleman HR, et al. Incidence and Survival of Primary Dermatofibrosarcoma Protuberans in the United States. *Dermatol Surg* 2016;42 Suppl 1:S24–31. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/26730971>.
4. Criscito MC, Martires KJ, Stein JA. Prognostic Factors, Treatment, and Survival in Dermatofibrosarcoma Protuberans. *JAMA Dermatol* 2016;152:1365–1371. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/27262160>.
5. Martin L, Piette F, Blanc P, et al. Clinical variants of the preprotuberant stage of dermatofibrosarcoma protuberans. *Br J Dermatol* 2005;153:932–936. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/16225602>.
6. Haycox CL, Odland PB, Olbricht SM, Casey B. Dermatofibrosarcoma protuberans (DFSP): growth characteristics based on tumor modeling and a review of cases treated with Mohs micrographic surgery. *Ann Plast Surg* 1997;38:246–251. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/9088462>.
7. Ratner D, Thomas CO, Johnson TM, et al. Mohs micrographic surgery for the treatment of dermatofibrosarcoma protuberans. Results of a multiinstitutional series with an analysis of the extent of microscopic spread. *J Am Acad Dermatol* 1997;37:600–613. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/9344201>.
8. Hayakawa K, Matsumoto S, Ae K, et al. Risk factors for distant metastasis of dermatofibrosarcoma protuberans. *J Orthop Traumatol* 2016;17:261–266. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/27289468>.
9. Vidimos AT, Helm TN, Papay FA. Dermatofibrosarcoma protuberans. In: *Cutaneous Oncology: Pathophysiology, Diagnosis, and Management*. Malden, MA: Blackwell Scientific; 1998.
10. PubMed Overview. Available at: <https://pubmed.ncbi.nlm.nih.gov/about/>.
11. Freedman-Cass DA, Fischer T, Alpert AB, et al. The Value and Process of Inclusion: Using Sensitive, Respectful, and Inclusive Language and Images in NCCN Content. *J Natl Compr Canc Netw* 2023;21:434–441. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/37156485>.
12. Mentzel T, Pedeutour F, Lazar A, Coindre JM, eds. Dermatofibrosarcoma Protuberans in WHO Classification of Tumors of Soft Tissue and Bone. In: Fletcher CD, Bridge JA, Hogendoorn PCW, Mertens F, eds (ed 4): IARC Press; 2013.
13. Connective Tissue Tumors. In: Calonje E, Brenn T, Lazar A, McKee PH, eds. *McKee's Pathology of the Skin with Clinical Correlations* (ed 4): Elsevier Saunders; 2012:1630–1635.
14. Abenoza P, Lillemoe T. CD34 and factor XIIIa in the differential diagnosis of dermatofibroma and dermatofibrosarcoma protuberans. *Am J Dermatopathol* 1993;15:429–434. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/7694515>.
15. Aiba S, Tabata N, Ishii H, et al. Dermatofibrosarcoma protuberans is a unique fibrohistiocytic tumour expressing CD34. *Br J Dermatol* 1992;127:79–84. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/1382538>.
16. Kutzner H. Expression of the human progenitor cell antigen CD34 (HPCA-1) distinguishes dermatofibrosarcoma protuberans from fibrous histiocytoma in formalin-fixed, paraffin-embedded tissue. *J Am Acad Dermatol* 1993;28:613–617. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/7681857>.
17. Goldblum JR, Tuthill RJ. CD34 and factor-XIIIa immunoreactivity in dermatofibrosarcoma protuberans and dermatofibroma. *Am J*

参考文献

1. Criscione VD, Weinstock MA. 美国1973年至2002年间隆起性皮肤纤维肉瘤的描述性流行病学研究. 美国皮肤病学杂志 2007; 56:968–973. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/17141362>.
2. Rouhani P, Fletcher CD, Devesa SS, Toro JR. 美国皮肤软组织肉瘤发病率模式：12,114例病例分析。Cancer 2008; 113:616–627. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/18618615>.
3. Kreicher KL, Kurlander DE, Gittleman HR, 等. 美国原发性隆起性皮肤纤维肉瘤的发病率与生存率. 《皮肤外科杂志》2016;42 Suppl 1: S24–31. 访问地址: <https://www.ncbi.nlm.nih.gov/pubmed/26730971>.
4. Criscito MC, Martires KJ, Stein JA. 凸出性皮肤纤维肉瘤的预后因素、治疗及生存率. JAMA Dermatol 2016; 152:1365–1371. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/27262160>.
5. Martin L, Piette F, Blanc P, 等. 隆起性皮肤纤维肉瘤隆起期前突起阶段的临床变异. 《英国皮肤病学杂志》2005;153:932–936. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/16225602>.
6. Haycox CL, Odland PB, Olbricht SM, Casey B. 隆起性皮肤纤维肉瘤 (DFSP)：基于肿瘤模型的生长特征及莫氏显微外科手术病例回顾. 《整形外科学年鉴》1997; 38:246–251. 可获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/9088462>.
7. Ratner D, Thomas CO, Johnson TM, 等学者. 莫氏显微外科技术治疗隆起性皮肤纤维肉瘤——一项多机构系列研究及显微镜下扩散范围分析结果. 《美国皮肤病学学会杂志》1997;37:600–613. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/9344201>.
8. Hayakawa K, Matsumoto S, Ae K 等. 凸出性皮肤纤维肉瘤远处转移的危险因素. 《骨科与创伤学杂志》2016; 17:261–266. 可获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/27289468>.
9. Vidimos AT, Helm TN, Papay FA. 凸出性皮肤纤维肉瘤. 载于: 《皮肤肿瘤学：病理生理学、诊断与治疗》。马萨诸塞州梅尔登: Blackwell Scientific; 1998年.
10. PubMed概览. 网址: <https://pubmed.ncbi.nlm.nih.gov/about/>.
11. Freedman-Cass DA, Fischer T, Alpert AB, 等. 包容的价值与实践:在NCCN内容中运用敏感、尊重且包容的语言与图像. 《美国国家综合癌症网络杂志》2023;21:434–441. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/37156485>.
12. Mentzel T, Pedetour F, Lazar A, Coindre JM (编). 《世界卫生组织软组织和骨骼肿瘤分类》中的隆起性皮肤纤维肉瘤. 见: Fletcher CD, Bridge JA, Hogendoorn PCW, Mertens F (编) (第4版): IARC出版社; 2013年.
13. 结缔组织肿瘤. 见: Calonje E, Brenn T, Lazar A, McKee PH (编). 《McKee皮肤病理学与临床相关性》(第4版): Elsevier Saunders出版社; 2012: 1630–1635.
14. Abenoza P, Lillemoe T. CD34与XIIIa因子在皮肤纤维瘤与隆起性皮肤纤维肉瘤鉴别诊断中的应用. 《美国皮肤病病理学杂志》1993; 15:429–434. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/7694515>.
15. Aiba S, Tabata N, Ishii H, 等学者指出,隆突性皮肤纤维纤维肉瘤是一种独特的纤维组织细胞肿瘤,可表达CD34. 《英国皮肤病学杂志》1992; 127:79–84. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/1382538>.
16. Kutzner H. 人类祖细胞抗原CD34 (HPCA-1) 的表达可区分福尔马林固定、石蜡包埋组织中的隆起性皮肤纤维肉瘤与纤维组织细胞瘤. 《美国皮肤病学学会杂志》1993; 28:613–617. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/7681857>.
17. Goldblum JR, Tuthill RJ. 凸出性皮肤纤维肉瘤与皮肤纤维瘤中的CD34及XIIIa因子免疫反应性. 《美国医学杂志》



Dermatopathol 1997;19:147–153. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/9129699>.

18. Sellheyer K, Nelson P, Krahl D. Dermatofibrosarcoma protuberans: a tumour of nestin-positive cutaneous mesenchymal stem cells? Br J Dermatol 2009;161:1317–1322. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/19659472>.

19. Cribier B, Noacco G, Peltre B, Grosshans E. Stromelysin 3 expression: a useful marker for the differential diagnosis dermatofibroma versus dermatofibrosarcoma protuberans. J Am Acad Dermatol 2002;46:408–413. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/11862177>.

20. Lisovsky M, Hoang MP, Dresser KA, et al. Apolipoprotein D in CD34-positive and CD34-negative cutaneous neoplasms: a useful marker in differentiating superficial acral fibromyxoma from dermatofibrosarcoma protuberans. Mod Pathol 2008;21:31–38. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/17885669>.

21. Yan X, Takahara M, Xie L, et al. Cathepsin K expression: a useful marker for the differential diagnosis of dermatofibroma and dermatofibrosarcoma protuberans. Histopathology 2010;57:486–488. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/20727023>.

22. Buck DW, 2nd, Kim JY, Alam M, et al. Multidisciplinary approach to the management of dermatofibrosarcoma protuberans. J Am Acad Dermatol 2012;67:861–866. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/22387033>.

23. Chappell AG, Doe SC, Worley B, et al. Multidisciplinary surgical treatment approach for dermatofibrosarcoma protuberans: an update. Arch Dermatol Res 2021;313:367–372. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/32770258>.

24. Curtis KK, Fakult NJ, Strunck JL, et al. Establishing Consensus for Mohs Micrographic Surgical Techniques in the Treatment of Melanoma in Situ for Future Clinical Trials: A Modified Delphi Study. J Natl Compr Canc Netw 2024;1–6. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/39079545>.

25. Connelly JH, Evans HL. Dermatofibrosarcoma protuberans. A clinicopathologic review with emphasis on fibrosarcomatous areas. Am J Surg Pathol 1992;16:921–925. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/1415902>.

26. Szollosi Z, Nemes Z. Transformed dermatofibrosarcoma protuberans: a clinicopathological study of eight cases. J Clin Pathol 2005;58:751–756. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/15976346>.

27. Mentzel T, Beham A, Katenkamp D, et al. Fibrosarcomatous ("high-grade") dermatofibrosarcoma protuberans: clinicopathologic and immunohistochemical study of a series of 41 cases with emphasis on prognostic significance. Am J Surg Pathol 1998;22:576–587. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/9591728>.

28. Bowne WB, Antonescu CR, Leung DH, et al. Dermatofibrosarcoma protuberans: A clinicopathologic analysis of patients treated and followed at a single institution. Cancer 2000;88:2711–2720. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/10870053>.

29. Abbott JJ, Oliveira AM, Nascimento AG. The prognostic significance of fibrosarcomatous transformation in dermatofibrosarcoma protuberans. Am J Surg Pathol 2006;30:436–443. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/16625088>.

30. Ding J, Hashimoto H, Enjoji M. Dermatofibrosarcoma protuberans with fibrosarcomatous areas. A clinicopathologic study of nine cases and a comparison with allied tumors. Cancer 1989;64:721–729. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/2545326>.

31. Hoesly PM, Lowe GC, Lohse CM, et al. Prognostic impact of fibrosarcomatous transformation in dermatofibrosarcoma protuberans: a cohort study. J Am Acad Dermatol 2015;72:419–425. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/25582537>.

32. Lyu A, Wang Q. Dermatofibrosarcoma protuberans: A clinical analysis. Oncol Lett 2018;16:1855–1862. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/30008876>.

33. Erdem O, Wyatt AJ, Lin E, et al. Dermatofibrosarcoma protuberans treated with wide local excision and followed at a cancer

Dermatopathol 1997; 19:147–153。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/9129699>。

18. Selheyer K, Nelson P, Krahl D. 凸出性皮肤纤维肉瘤：一种巢蛋白阳性皮肤间充质干细胞肿瘤？《英国皮肤病学杂志》2009; 161:1317–1322。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/19659472>。

19. Cribier B, Noacco G, Peltre B, Grosshans E. Stromelysin 3 表达：区分皮肤纤维瘤与隆起性皮肤纤维肉瘤的有效标志物。《美国皮肤病学杂志》2002; 46:408–413。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/11862177>。

20. Lisovsky M, Hoang MP, Dresser KA, 等. 载脂蛋白D在CD34阳性与CD34阴性皮肤肿瘤中的应用：区分浅表肢端纤维黏液瘤与隆起性皮肤纤维肉瘤的有效标志物。《病理学与病理生理学杂志》2008;21:31–38。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/17885669>。

21. Yan X, Takahara M, Xie L, 等. 组织蛋白酶K表达：皮肤纤维瘤与隆起性皮肤纤维肉瘤鉴别诊断的有效标志物。《组织病理学》2010;57:486–488。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/20727023>。

22. Buck DW, 2nd, Kim JY, Alam M, 等. 采用多学科方法治疗隆起性皮肤纤维肉瘤。《美国皮肤病学会杂志》2012; 67:861–866。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/22387033>。

23. Chappell AG, Doe SC, Worley B, et al. 皮肤纤维肉瘤隆起型的多学科手术治疗方案：最新进展。Arch Dermatol Res 2021;313:367–372。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/32770258>。

24. Curtis KK, Fakult NJ, Strunck JL, 等学者：为未来临床试验确立原位黑色素瘤莫氏显微外科技术的共识：一项改良德尔菲研究。《美国国家综合癌症网络杂志》2024年：1–6。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/39079545>。

25. Connelly JH, Evans HL. 凸出性皮肤纤维肉瘤。一项侧重于纤维肉瘤区域的临床病理学综述。《美国外科病理学杂志》1992; 16:921–925。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/1415902>。

26. Szollosi Z, Nemes Z. 变形性皮肤纤维肉瘤隆起型：八例临床病理学研究。《临床病理学杂志》2005; 58:751–756。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/15976346>。

27. Mentzel T, Beham A, Katenkamp D 等. 纤维肉瘤样（“高级别”）隆起性皮肤纤维肉瘤：41例病例系列的临床病理学与免疫组织化学研究，重点探讨其预后意义。Am J Surg Pathol 1998; 22:576–587。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/9591728>。

28. Bowne WB, Antonescu CR, Leung DH, 等. 隆起性皮肤纤维肉瘤：单一机构收治及随访患者的临床病理学分析。《癌症》2000;88:2711–2720。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/10870053>。

29. Abbott JJ, Oliveira AM, Nascimento AG. 皮肤纤维肉瘤性隆起瘤中纤维肉瘤样转化的预后意义。Am J Surg Pathol 2006; 30:436–443。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/16625088>。

30. 丁J、桥本H、恩乔吉M. 伴纤维肉瘤样区域的隆起性皮肤纤维肉瘤：九例临床病理学研究及与相关肿瘤的比较。《癌症》1989; 64:721–729。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/2545326>。

31. Hoesly PM, Lowe GC, Lohse CM, 等. 隆起性皮肤纤维肉瘤中纤维肉瘤样转化的预后影响：一项队列研究。《美国皮肤病学杂志》2015;72:419–425。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/25582537>。

32. Lyu A, Wang Q. 凸出性皮肤纤维肉瘤：一项临床分析。Oncol Lett 2018; 16:1855–1862。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/30008876>。

33. Erdem O, Wyatt AJ, Lin E 等. 采用广泛局部切除术治疗的隆起性皮肤纤维肉瘤，并在癌症专科医院进行随访。



hospital: prognostic significance of clinicopathologic variables. *Am J Dermatopathol* 2012;34:24–34. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/21785324>.

34. Jing C, Zhang H, Zhang X, Yu S. Dermatofibrosarcoma Protuberans: A Clinicopathologic and Therapeutic Analysis of 254 Cases at a Single Institution. *Dermatol Surg* 2021;47:e26–e30. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/32769521>.

35. Li Y, Wang C, Xiang B, et al. Clinical Features, Pathological Findings and Treatment of Recurrent Dermatofibrosarcoma Protuberans. *J Cancer* 2017;8:1319–1323. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/28607608>.

36. Goldblum JR, Reith JD, Weiss SW. Sarcomas arising in dermatofibrosarcoma protuberans: a reappraisal of biologic behavior in eighteen cases treated by wide local excision with extended clinical follow up. *Am J Surg Pathol* 2000;24:1125–1130. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/10935653>.

37. Castle KO, Guadagnolo BA, Tsai CJ, et al. Dermatofibrosarcoma protuberans: long-term outcomes of 53 patients treated with conservative surgery and radiation therapy. *Int J Radiat Oncol Biol Phys* 2013;86:585–590. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/23628134>.

38. Liang CA, Jambusaria-Pahlajani A, Karia PS, et al. A systematic review of outcome data for dermatofibrosarcoma protuberans with and without fibrosarcomatous change. *J Am Acad Dermatol* 2014;71:781–786. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/24755121>.

39. Cai H, Wang Y, Wu J, Shi Y. Dermatofibrosarcoma protuberans: clinical diagnoses and treatment results of 260 cases in China. *J Surg Oncol* 2012;105:142–148. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/21815146>.

40. Ugurel S, Mentzel T, Utikal J, et al. Neoadjuvant imatinib in advanced primary or locally recurrent dermatofibrosarcoma protuberans: a multicenter phase II DeCOG trial with long-term follow-up. *Clin Cancer Res* 2014;20:499–510. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/24173542>.

41. Zhou X, Sun D, Liu Y, et al. Dermatofibrosarcoma protuberans: our 10-year experience on 80 patients. *J Dermatolog Treat* 2020;31:554–558. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/31116621>.

42. Kimmel Z, Ratner D, Kim JY, et al. Peripheral excision margins for dermatofibrosarcoma protuberans: a meta-analysis of spatial data. *Ann Surg Oncol* 2007;14:2113–2120. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/17468914>.

43. Chen Y, Jiang G. Association between surgical excision margins and outcomes in patients with dermatofibrosarcoma protuberans: A meta-analysis. *Dermatol Ther* 2021;34:e14954. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/33835635>.

44. Gloster HM, Jr., Harris KR, Roenigk RK. A comparison between Mohs micrographic surgery and wide surgical excision for the treatment of dermatofibrosarcoma protuberans. *J Am Acad Dermatol* 1996;35:82–87. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/8682970>.

45. DuBay D, Cimmino V, Lowe L, et al. Low recurrence rate after surgery for dermatofibrosarcoma protuberans: a multidisciplinary approach from a single institution. *Cancer* 2004;100:1008–1016. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/14983497>.

46. Meguerditchian AN, Wang J, Lema B, et al. Wide excision or Mohs micrographic surgery for the treatment of primary dermatofibrosarcoma protuberans. *Am J Clin Oncol* 2010;33:300–303. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/19858696>.

47. Foroozan M, Sei JF, Amini M, et al. Efficacy of Mohs micrographic surgery for the treatment of dermatofibrosarcoma protuberans: systematic review. *Arch Dermatol* 2012;148:1055–1063. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/22986859>.

48. Bogucki B, Neuhaus I, Hurst EA. Dermatofibrosarcoma protuberans: a review of the literature. *Dermatol Surg* 2012;38:537–551. Available at:

<https://www.ncbi.nlm.nih.gov/pubmed/22288484>.

49. Paradisi A, Abeni D, Rusconi A, et al. Dermatofibrosarcoma protuberans: wide local excision vs. Mohs micrographic surgery.

医院：临床病理学变量的预后意义。《美国皮肤病病理学杂志》2012；34:24–34。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/21785324>。

34. Jing C, Zhang H, Zhang X, Yu S. 凸出性皮肤纤维肉瘤：单一机构254例病例的临床病理学与治疗分析。Dermatol Surg 2021；47:e26–e30。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/32769521>。

35. Li Y, Wang C, Xiang B, 等.复发性隆起性皮肤纤维肉瘤的临床特征，病理学发现及治疗。《癌症杂志》2017；8:1319–1323。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/28607608>。

36. Goldblum JR, Reith JD, Weiss SW. 由隆起性皮肤纤维肉瘤发生的肉瘤：通过对18例患者行广泛局部切除术并进行长期临床随访后对其生物学行为的重新评估。Am J Surg Pathol 2000；24:1125–1130。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/10935653>。

37. Castle KO, Guadagnolo BA, Tsai CJ, 等.隆突性皮肤纤维肉瘤:53例接受保守手术联合放射治疗患者的长期预后。《国际放射肿瘤学与生物物理学杂志》2013；86:585–590。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/23628134>。

38. 梁CA、詹布萨里亚-帕拉贾尼A、卡里亚PS等。关于伴或不伴纤维肉瘤样改变的隆起性皮肤纤维肉瘤结局数据的系统综述。《美国皮肤病学杂志》2014；71:781–786。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/24755121>。

39. 蔡H、王Y、吴J、史Y。隆突性皮肤纤维肉瘤：中国260例临床诊断与治疗结果。《外科肿瘤学杂志》2012；105:142–148。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/21815146>。

40. Ugurel S, Mentzel T, Utikal J, 等.新辅助伊马替尼治疗晚期原发性或局部复发性隆起型皮肤纤维肉瘤:一项多中心II期DeCOG试验及长期随访。《临床癌症研究》2014；20:499–510。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/24173542>。

41. Zhou X, Sun D, Liu Y, 等.隆突性皮肤纤维肉瘤:我们对80例患者的10年经验。《皮肤病学治疗杂志》2020；31:554–558。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/31116621>。

42. Kimmel Z, Ratner D, Kim JY, 等.隆起性皮肤纤维肉瘤的外周切除边缘:一项空间数据的荟萃分析。《外科肿瘤学年鉴》2007；14:2113–2120。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/17468914>。

43. 陈Y, 江G。隆起性皮肤纤维肉瘤患者手术切除边缘与预后之间的关联：一项荟萃分析。《皮肤科治疗》2021；34:e14954。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/33835635>。

44. Gloster HM, Jr., Harris KR, Roenigk RK. Mohs显微外科手术与广泛手术切除治疗隆起性皮肤纤维肉瘤的比较。《美国皮肤病学会杂志》1996；35:82–87。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/8682970>。

45. DuBay D, Cimmino V, Lowe L, 等.隆起性皮肤纤维肉瘤术后低复发率:来自单一机构的多学科诊疗方案。《癌症》2004；100:1008–1016。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/14983497>。

46. Meguerditchian AN, Wang J, Lema B, 等.广泛切除术或莫氏显微外科手术治疗原发性隆起性皮肤纤维肉瘤。《美国临床肿瘤学杂志》2010；33:300–303。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/19858696>。

47. Foroozan M, Sei JF, Amini M, 等.莫氏显微外科手术治疗隆起性皮肤纤维肉瘤的疗效:系统评价。《皮肤病学档案》2012；148:1055–1063。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/22986859>。

48. Bogucki B, Neuhaus I, Hurst EA. 凸出性皮肤纤维肉瘤：文献综述。Dermatol Surg 2012；38:537–

551。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/22288484>。

49. Paradisi A, Abeni D, Rusconi A, 等学者:隆起性皮肤纤维肉瘤:广泛局部切除术与莫氏显微外科手术的比较。



Cancer Treat Rev 2008;34:728–736. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/18684568>.

50. Lowe GC, Onajin O, Baum CL, et al. A comparison of mohs micrographic surgery and wide local excision for treatment of dermatofibrosarcoma protuberans with long-term follow-up: The mayo clinic experience. *Dermatol Surg* 2017;43:98–106. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/27749444>.

51. Veronese F, Boggio P, Tiberio R, et al. Wide local excision vs. Mohs Tubingen technique in the treatment of dermatofibrosarcoma protuberans: a two-centre retrospective study and literature review. *J Eur Acad Dermatol Venereol* 2017;31:2069–2076. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/28573714>.

52. Malan M, Xuejingzi W, Quan SJ. The efficacy of Mohs micrographic surgery over the traditional wide local excision surgery in the cure of dermatofibrosarcoma protuberans. *Pan Afr Med J* 2019;33:297. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/31692830>.

53. Durack A, Gran S, Gardiner MD, et al. A 10-year review of surgical management of dermatofibrosarcoma protuberans. *Br J Dermatol* 2021;184:731–739. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/32599647>.

54. Farma JM, Ammori JB, Zager JS, et al. Dermatofibrosarcoma protuberans: how wide should we resect? *Ann Surg Oncol* 2010;17:2112–2118. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/20354798>.

55. Snow H, Davies E, Strauss DC, et al. Conservative re-excision is a safe and simple alternative to radical resection in revision surgery for dermatofibrosarcoma protuberans. *Ann Surg Oncol* 2020;27:919–923. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/31664620>.

56. Harati K, Lange K, Goertz O, et al. A single-institutional review of 68 patients with dermatofibrosarcoma protuberans: wide re-excision after inadequate previous surgery results in a high rate of local control. *World J Surg Oncol* 2017;15:5. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/28056985>.

57. Suit H, Spiro I, Mankin HJ, et al. Radiation in management of patients with dermatofibrosarcoma protuberans. *J Clin Oncol* 1996;14:2365–2369. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/8708729>.

58. Uysal B, Sager O, Gamsiz H, et al. Evaluation of the role of radiotherapy in the management of dermatofibrosarcoma protuberans. *J BUON* 2013;18:268–273. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/23613415>.

59. Hamid R, Hafeez A, Darzi MA, et al. Outcome of wide local excision in dermatofibrosarcoma protuberans and use of radiotherapy for margin-positive disease. *Indian Dermatol Online J* 2013;4:93–96. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/23741663>.

60. Ballo MT, Zagars GK, Pisters P, Pollack A. The role of radiation therapy in the management of dermatofibrosarcoma protuberans. *Int J Radiat Oncol Biol Phys* 1998;40:823–827. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/9531366>.

61. Dagan R, Morris CG, Zlotecki RA, et al. Radiotherapy in the treatment of dermatofibrosarcoma protuberans. *Am J Clin Oncol* 2005;28:537–539. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/16317260>.

62. Sun LM, Wang CJ, Huang CC, et al. Dermatofibrosarcoma protuberans: treatment results of 35 cases. *Radiother Oncol* 2000;57:175–181. Available at:
<https://www.ncbi.nlm.nih.gov/pubmed/11054521>.

63. Williams N, Morris CG, Kirwan JM, et al. Radiotherapy for dermatofibrosarcoma protuberans. *Am J Clin Oncol* 2014;37:430–432. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/23388563>.

64. Haas RL, Keus RB, Loftus BM, et al. The role of radiotherapy in the local management of dermatofibrosarcoma protuberans. *Soft Tissue Tumours Working Group. Eur J Cancer* 1997;33:1055–1060. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/9376187>.

65. Chen YT, Tu WT, Lee WR, Huang YC. The efficacy of adjuvant radiotherapy in dermatofibrosarcoma protuberans: a systemic review

《癌症治疗评论》2008年第34卷：728–736页。获取地

址：<https://www.ncbi.nlm.nih.gov/pubmed/18684568>。

50. Lowe GC, Onajin O, Baum CL, et al. Mohs显微外科手术与广泛局部切除术治疗隆起性皮肤纤维肉瘤的长期随访比较:梅奥诊所经验. *Dermatol Surg* 2017;43:98–106. 获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/27749444>.

51. Veronese F, Boggio P, Tiberio R, 等学者:广泛局部切除术与T-莫氏术式治疗隆起性皮肤纤维肉瘤的比较:一项双中心回顾性研究及文献综述.《欧洲皮肤病学与性病学会杂志》2017;31:2069–2076.获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/28573714>.

52. Malan M, Xuejingzi W, Quan SJ. Mohs显微外科术相较于传统广泛局部切除术治疗隆起性皮肤纤维肉瘤的疗效. *Pan Afr Med J* 2019;33:297. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/31692830>。

53. Durack A, Gran S, Gardiner MD, 等.隆起性皮肤纤维肉瘤外科治疗的10年回顾.《英国皮肤病学杂志》2021;184:731–739.获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/32599647>.

54. Farma JM, Ammori JB, Zager JS, 等学者.隆突性皮肤纤维肉瘤:应切除多广泛?《外科肿瘤学年鉴》2010;17:2112–2118.获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/20354798>.

55. Snow H, Davies E, Strauss DC, 等学者指出:对于隆起性皮肤纤维肉瘤的修正手术,保守性再次切除术是根治性切除术的一种安全且简便的替代方案.《外科肿瘤学年鉴》2020;27:919–923.获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/31664620>.

56. Harati K, Lange K, Goertz O 等人.一项单机构综述
68例隆起性皮肤纤维肉瘤患者:在既往手术效果不佳的情况下进行广泛再次切除可实现较高的局部控制率.《世界外科肿瘤学杂志》2017;15:5.获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/28056985>。

57. Suit H, Spiro I, Mankin HJ, 等.放射疗法在隆起性皮肤纤维肉瘤患者治疗中的应用.《临床肿瘤学杂志》1996;14:2365–2369.获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/8708729>.

58. Uysal B, Sager O, Gamsiz H, 等.放射治疗在隆起性皮肤纤维肉瘤治疗中的作用评估. *J BUON* 2013;18:268–273. 参考来源:<https://www.ncbi.nlm.nih.gov/pubmed/23613415>.

59. Hamid R, Hafeez A, Darzi MA, 等.隆起型皮肤纤维肉瘤广泛局部切除术的疗效及边缘阳性病变的放射治疗应用.《印度皮肤病学在线杂志》2013;4:93–96. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/23741663>.

60. Ballo MT, Zagars GK, Pisters P, Pollack A. 放射治疗在隆突性皮肤纤维肉瘤治疗中的作用.《国际放射肿瘤学与生物物理学杂志》1998;40:823–827. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/9531366>。

61. Dagan R, Morris CG, Zlotecki RA, 等.放射治疗在隆突性皮肤纤维肉瘤治疗中的应用.《美国临床肿瘤学杂志》2005;28:537–539.获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/16317260>.

62. Sun LM, Wang CJ, Huang CC, 等.隆突性皮肤纤维肉瘤:35例治疗结果.《放射肿瘤学》2000;57:175–181. 获取地址: <https://www.ncbi.nlm.nih.gov/pubmed/11054521>.

63. Williams N, Morris CG, Kirwan JM, 等.隆突性皮肤纤维肉瘤的放射治疗.《美国临床肿瘤学杂志》2014;37:430–432.获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/23388563>.

64. Haas RL, Keus RB, Loftus BM, 等.放射治疗在隆起性皮肤纤维肉瘤局部治疗中的作用.软组织肿瘤工作组.《欧洲癌症杂志》1997;33:1055–1060.获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/9376187>.

65. 陈YT、屠WT、李WR、黄YC.辅助放疗在隆起性皮肤纤维肉瘤中的疗效:系统性综述



and meta-analysis. J Eur Acad Dermatol Venereol 2016;30:1107–1114. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/26879523>.

66. Takahira T, Oda Y, Tamiya S, et al. Detection of COL1A1-PDGFB fusion transcripts and PDGFB/PDGFRB mRNA expression in dermatofibrosarcoma protuberans. Mod Pathol 2007;20:668–675. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/17431412>.

67. Patel KU, Szabo SS, Hernandez VS, et al. Dermatofibrosarcoma protuberans COL1A1-PDGFB fusion is identified in virtually all dermatofibrosarcoma protuberans cases when investigated by newly developed multiplex reverse transcription polymerase chain reaction and fluorescence in situ hybridization assays. Hum Pathol 2008;39:184–193. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/17950782>.

68. Simon MP, Pedeutour F, Sirvent N, et al. Deregulation of the platelet-derived growth factor B-chain gene via fusion with collagen gene COL1A1 in dermatofibrosarcoma protuberans and giant-cell fibroblastoma. Nat Genet 1997;15:95–98. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/8988177>.

69. Rutkowski P, Van Glabbeke M, Rankin CJ, et al. Imatinib mesylate in advanced dermatofibrosarcoma protuberans: pooled analysis of two phase II clinical trials. J Clin Oncol 2010;28:1772–1779. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/20194851>.

70. Navarrete-Dechent C, Mori S, Barker CA, et al. Imatinib treatment for locally advanced or metastatic dermatofibrosarcoma protuberans: A systematic review. JAMA Dermatol 2019;155:361–369. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/30601909>.

71. Kerob D, Porcher R, Verola O, et al. Imatinib mesylate as a preoperative therapy in dermatofibrosarcoma: results of a multicenter phase II study on 25 patients. Clin Cancer Res 2010;16:3288–3295. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/20439456>.

72. McArthur GA, Demetri GD, van Oosterom A, et al. Molecular and clinical analysis of locally advanced dermatofibrosarcoma protuberans treated with imatinib: Imatinib Target Exploration Consortium Study B2225. J Clin Oncol 2005;23:866–873. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/15681532>.

73. Rutkowski P, Debiec-Rychter M, Nowecki Z, et al. Treatment of advanced dermatofibrosarcoma protuberans with imatinib mesylate with or without surgical resection. J Eur Acad Dermatol Venereol 2011;25:264–270. Available at: <https://www.ncbi.nlm.nih.gov/pubmed/20569296>.

及荟萃分析。《欧洲皮肤病学与性病学学会杂志》2016; 30:1107–1114。获取地址：<https://www.ncbi.nlm.nih.gov/pubmed/26879523>。

66. Takahira T, Oda Y, Tamiya S, 等. 皮肤纤维肉瘤隆起型中COL1A1-PDGFB融合转录本及PDGFB/PDGFRB mRNA表达的检测. Mod Pathol 2007;20:668–675. 参见:<https://www.ncbi.nlm.nih.gov/pubmed/17431412>.

67. Patel KU, Szabo SS, Hernandez VS, 等人通过新开发的多重逆转录聚合酶链反应和荧光原位杂交技术检测,几乎在所有凸性皮肤纤维肉瘤病例中均能检测到COL1A1-PDGFB融合基因.n项检测.《人类病理学》2008;39:184–193. 获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/17950782>.

68. Simon MP, Pedeutour F, Sirvent N, 等.在隆起性皮肤纤维肉瘤和巨细胞纤维母细胞瘤中,通过与胶原基因COL1A1融合导致血小板衍生生长因子B链基因的失调.《自然·遗传学》1997;15:95–98.获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/8988177>.

69. Rutkowski P, Van Glabbeke M, Rankin CJ, 等.甲磺酸伊马替尼治疗晚期隆起型皮肤纤维肉瘤:两项II期临床试验的汇总分析.《临床肿瘤学杂志》2010;28:1772–1779. 获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/20194851>.

70. Navarrete-Dechent C, Mori S, Barker CA, 等.伊马替尼治疗局部晚期或转移性隆起性皮肤纤维肉瘤:一项系统性综述.JAMA Dermatol 2019; 155:361–369.获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/30601909>.

71. Kerob D, Porcher R, Verola O, 等.甲磺酸伊马替尼作为皮肤纤维肉瘤的术前治疗:一项针对25例患者的多中心II期研究结果.《临床癌症研究》2010;16:3288–3295. 获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/20439456>.

72. McArthur GA, Demetri GD, van Oosterom A, 等.伊马替尼治疗局部晚期隆起型皮肤纤维肉瘤的分子与临床分析:伊马替尼靶点探索联盟研究B 2225.《临床肿瘤学杂志》2005;23:866–873. 参考来源:<https://www.ncbi.nlm.nih.gov/pubmed/15681532>.

73. Rutkowski P, Debiec-Rychter M, Nowecki Z, 等.采用甲磺酸伊马替尼联合或不联合手术切除治疗晚期隆起性皮肤纤维肉瘤.《欧洲皮肤病学与性病学学会杂志》2011;25:264–270.可获取地址:<https://www.ncbi.nlm.nih.gov/pubmed/20569296>。